

SECTION A

Select the best single answer

1. For general clinical descriptions, which of the following planes is used as one of the planes to define four quadrants of the abdominal cavity?
(A) subcostal
(B) transtubercular
(C) midclavicular
(D) transumbilical
(E) midaxillary
2. The midclavicular planes pass through the midpoint of the clavicles to the midpoint of which of the following structures?
(A) anterior superior iliac spine
(B) symphysis pubis
(C) umbilicus
(D) inguinal ligament
(E) xiphoid process
3. A retroperitoneal mass in the lower right quadrant is putting pressure on a portion of the gastrointestinal tract, causing digestive issues and discomfort. Which of the following structures may be affected by the pressure of the mass?
(A) Sigmoid colon
(B) Rectum
(C) Jejunum
(D) Duodenum
(E) Cecum
4. A 31-year-old male patient is admitted to the emergency department after suffering multiple injuries in a motor vehicle collision. He is found to have a lacerated liver and the surgical team is consulted and prepped for emergency surgery to repair the organ. Surgeons found that the liver was lacerated anteriorly between the gallbladder and round ligament (ligamentum teres hepatis). Which of the following liver structures is located in the region of the laceration?
A. Bare area
B. Left lobe
C. Caudate lobe
D. Ductus venosus
E. Quadrate lobe
5. Which of the following is NOT a type of lingual papilla?
(A) vallate papillae
(B) foliate papillae
(C) bacilliform papillae
(D) filiform papillae
(E) fungiform papillae

6. The anterolateral abdominal wall is bounded by all of the following structures EXCEPT
- (A) cartilages of the 7th through 10th ribs
 - (B) linea alba
 - (C) xiphoid process
 - (D) inguinal ligament
 - (E) pelvic bone
7. Which of the following muscles is NOT innervated by the hypoglossal nerve?
- (A) genioglossus
 - (B) hyoglossus
 - (C) styloglossus
 - (D) palatoglossus
 - (E) intrinsic muscles of the tongue
8. All of the following structures are located within the rectus sheath EXCEPT
- (A) pyramidalis
 - (B) rectus abdominis
 - (C) inferior epigastric arteries and veins
 - (D) deep inguinal ring
 - (E) ventral primary rami of T7–T12 nerves
9. Inferiorly, the inferior margin of the external oblique aponeurosis thickens and folds back on itself to form which of the following structures?
- (A) rectus sheath
 - (B) inguinal ligament
 - (C) arcuate line
 - (D) deep inguinal ring
 - (E) fundiform ligament
10. The rectus abdominis muscle is anchored transversely by attachment to the anterior layer of the rectus sheath by which of the following structures?
- (A) pubic tubercle
 - (B) xiphoid process
 - (C) linea alba
 - (D) tendinous intersections
 - (E) umbilicus
11. The two medial umbilical folds represent remnants of which of the following structures?
- (A) urachus
 - (B) umbilical arteries
 - (C) umbilical veins
 - (D) ductus venosus
 - (E) ductus arteriosus
12. Which of the following structures give rise to the deep inguinal ring?
- (A) gubernaculum
 - (B) conjoint tendon
 - (C) lacunar ligament
 - (D) external abdominal oblique aponeurosis

(E) transversalis fascia

13. The peritoneal space contains which of the following?

- (A) liver
- (B) pancreas
- (C) large intestine
- (D) kidney
- (E) peritoneal fluid

14. Which of the following structures connects the lesser curvature of the stomach and the proximal part of the duodenum to the liver?

- (A) greater omentum
- (B) peritoneal ligament
- (C) mesentery
- (D) lesser omentum
- (E) peritoneal fold

15. Which of the following structures is often referred to as the "abdominal policeman"?

- (A) hepatoduodenal ligament
- (B) gastrohepatic ligament
- (C) greater omentum
- (D) gastrocolic ligament
- (E) falciform ligament

16. A 4-day-old boy has not defecated since coming home from the hospital even though his feeding pattern has been normal without any excessive vomiting. Rectal examination reveals a normal anus, anal canal, and rectum. However, a large faecal mass is found in the colon, and a large release of flatus and faeces follows the rectal examination. Which of the following conditions would be suspected?

- (A) Imperforate anus
- (B) Anal agenesis
- (C) Anorectal agenesis
- (D) Rectal atresia
- (E) Hirschsprung disease

17. Which of the following structures contains both smooth and skeletal muscles?

- (A) stomach
- (B) jejunum
- (C) cecum
- (D) esophagus
- (E) rectum

18. Which of the following arteries provides the abdominal parts of the oesophagus with its arterial supply?

- (A) cystic
- (B) gastroduodenal
- (C) left gastric
- (D) hepatic
- (E) left gastroepiploic

19. Rugae are located in which of the following structures?
(A) duodenum
(B) stomach
(C) cecum
(D) ileum
(E) transverse colon
20. The first part of the duodenum is located at which of the following vertebral levels?
(A) T10
(B) L2
(C) L1
(D) L5
(E) L3
21. The sympathetic nerve supply of the stomach arises from which of the following cord segments?
(A) T1–T5
(B) T6–T9
(C) T6–T12
(D) L1–L3
(E) T10–L2
22. The bile and pancreatic ducts enter which of the following structures?
(A) stomach
(B) 2nd portion of the duodenum
(C) cecum
(D) ileum
(E) liver
23. Which of the following structures crosses over the inferior or horizontal (third) portion of the duodenum?
(A) pancreas
(B) hepatic artery
(C) portal vein
(D) superior mesenteric artery
24. The duodenojejunal junction is supported by the attachment of which of the following structures?
(A) suspensory muscle of the duodenum (ligament of Treitz)
(B) falciform ligament
(C) hepatoduodenal ligament
(D) greater omentum
(E) transverse mesocolon
25. The superior anterior and posterior pancreaticoduodenal arteries arise from which of the following arteries?
(A) right colic
(B) ileocolic

- (C) gastroduodenal
- (D) hepatic
- (E) splenic

26. The superior mesenteric and splenic veins unite to form the portal vein posterior to which of the following structures?

- (A) right kidney
- (B) neck of the pancreas
- (C) pylorus of stomach
- (D) 2nd portion of the duodenum
- (E) spleen

27. Circular folds (plicae circulares) are characteristic of which of the following structures?

- (A) transverse colon
- (B) stomach
- (C) jejunum
- (D) cecum
- (E) sigmoid colon

28. Omental appendices are located on which of the following structures?

- (A) stomach
- (B) duodenum
- (C) ileum
- (D) ascending colon
- (E) liver

29. There are no teniae coli in which of the following structures?

- (A) ascending colon
- (B) transverse colon
- (C) descending colon
- (D) sigmoid colon
- (E) appendix

30. Which of the following structures can be located deep to a point that is one-third of the way along the oblique line, joining the right anterior superior iliac spine to the umbilicus?

- (A) gallbladder
- (B) spleen
- (C) right kidney
- (D) appendix
- (E) urinary bladder

31. The splenic artery usually follows a tortuous course along which of the following structures?

- (A) left kidney
- (B) greater curvature of the stomach
- (C) pancreas
- (D) transverse colon
- (E) cecum

32. The head of the pancreas is embraced by which of the following structures?

- (A) stomach
- (B) spleen
- (C) cecum
- (D) C-shaped curve of the duodenum
- (E) transverse mesocolon

33. The main pancreatic duct and the bile duct unite to form which of the following structures?

- (A) common bile duct
- (B) hepatic duct
- (C) accessory pancreatic duct
- (D) cystic duct
- (E) hepatopancreatic ampulla

34. The porta hepatis gives passage to all of the following structures EXCEPT

- (A) portal vein
- (B) hepatic artery
- (C) hepatic ducts
- (D) lymphatic vessels
- (E) cystic artery

35. The portion of the hepatic artery extending between the celiac trunk and the gastroduodenal artery is known as the

- (A) proper hepatic
- (B) common hepatic
- (C) right hepatic
- (D) left hepatic
- (E) middle hepatic

36. The hepatic veins drain into which of the following structures?

- (A) liver
- (B) inferior vena cava
- (C) spleen
- (D) portal vein
- (E) superior vena cava

37. The spiral valve is located in which of the following structures?

- (A) head of pancreas
- (B) pylorus
- (C) neck of gallbladder
- (D) cecum
- (E) rectum

38. The cystic artery commonly arises from the right hepatic artery in the angle between the common hepatic duct and which of the following structures?

- (A) cystic duct
- (B) celiac trunk
- (C) portal vein

- (D) proper hepatic artery
- (E) gastroduodenal artery

39. Which of the following veins, when dilated, produces caput medusae?

- (A) proper hepatic
- (B) splenic
- (C) cystic
- (D) paraumbilical
- (E) rectal

40. Inferiorly, the posterior surfaces of the kidney are related to all of the following structures EXCEPT

- (A) subcostal nerve
- (B) iliohypogastric nerve
- (C) ilioinguinal nerve
- (D) 2nd portion of the duodenum
- (E) quadratus lumborum

41. Following involvement in a car crash, a 20-year-old undergoes a splenectomy due to a lacerated spleen secondary to blunt abdominal trauma. During the procedure, the surgeon ligates the blood supply of the spleen at its hilum. Improper placement of the ligature may lead to damage to the arterial supply of which of the following structures?

- (A) Fundic portion of the stomach
- (B) Kidney
- (C) Pyloric portion of stomach
- (D) Head of pancreas
- (E) Splenic flexure of colon

42. The patient reports the bulge becomes more prominent when he is standing up, defecating, and coughing. On examination, the doctor places his finger in the location of the bulge and notes that his finger feels like it can pass into the abdominal cavity. Given his presentation and history, what is the most likely diagnosis?

- (A) Direct inguinal hernia
- (B) Indirect inguinal hernia
- (C) Femoral hernia
- (D) Lymphadenitis of superficial inguinal nodes
- (E) Hydrocele testis

43. The inferior vena cava begins anterior to which of the following structures?

- (A) right crus of the diaphragm
- (B) right kidney
- (C) 5th lumbar vertebra
- (D) crest of the ilium
- (E) bifurcation of the aorta

44. The anatomical left and right lobes of the liver are separated on the diaphragmatic surface of the liver by which of the following structures?

- (A) fissure for the round ligament of the liver

- (B) fissure for the ligamentum venosum
 - (C) falciform ligament
 - (D) porta hepatis
 - (E) lesser omentum
45. All of the following statements concerning the spleen are correct EXCEPT
- (A) It is the largest branch of the celiac trunk.
 - (B) It follows a tortuous course along the inferior border of the pancreas.
 - (C) It divides into five or more branches that enter the hilum of the spleen.
 - (D) It runs anterior to the left kidney.
 - (E) It follows posterior to the omental bursa.
46. In the developing embryo, the midgut rotates 270 degrees around which of the following structures?
- (A) superior mesenteric artery
 - (B) celiac trunk
 - (C) splenic artery
 - (D) left renal artery
 - (E) inferior vena cava
47. Which of the following statements correctly relates to the 3rd part of the duodenum?
- (A) It is supported by the suspensory ligament of the duodenum.
 - (B) The bile and pancreatic ducts enter its posteromedial wall.
 - (C) It is crossed by the superior mesenteric artery and vein.
 - (D) It attaches to the hepatoduodenal ligament.
 - (E) It lies to the right and runs parallel to the inferior vena cava.
48. Digestion occurs mainly in which of the following structures?
- (A) cecum and ascending colon
 - (B) transverse colon
 - (C) stomach and duodenum
 - (D) jejunum and ileum
 - (E) transverse and sigmoid colon
49. Most reabsorption of water occurs in which of the following structures?
- (A) stomach
 - (B) jejunum
 - (C) sigmoid colon
 - (D) ascending colon
 - (E) rectum
50. The bifurcation of the abdominal aorta occurs at the level of which of the following structures?
- (A) crest of the ilium
 - (B) inguinal ligament
 - (C) pubic tubercle
 - (D) symphysis pubis

(E) obturator foramen

51. Which of the following organs may be directly affected by malrotation of the caudal limb of the primary intestinal loop?

- (A) Descending colon
- (B) Stomach
- (C) Spleen
- (D) Fourth part of the duodenum
- (E) Ascending colon

52. During a splenectomy, a surgeon carefully dissects the peritoneal ligament containing the splenic vessels at the hilum of the spleen. Which of the following peritoneal ligaments is being dissected by the surgeon?

- (A) Hepatoduodenal ligament
- (B) Coronary ligament
- (C) Transverse mesocolon
- (D) Lienorenal ligament
- (E) Gastrosplenic ligament

53. Select the *incorrect statement* about the lips:

- (A) They bound the orifice of mouth
- (B) They are lined externally by the skin and internally by the mucous membrane
- (C) Their red portions are called vermillion zones
- (D) Lymphatics from lateral part of lower lip drain into submental lymph nodes

54. Which of the following types of teeth are not found in deciduous dentition?

- (A) Incisors
- (B) Canines
- (C) Premolars
- (D) Molars

55. Select the *incorrect statement* about the root of the tongue:

- (A) It is the part of tongue that rests on the floor of the mouth
- (B) It is attached by the extrinsic muscles to the mandible and hyoid bone
- (C) The nerves and vessels of tongue enter or leave through the root
- (D) It is attached by the intrinsic muscles to the mandible and hyoid bone

56. All the muscles of the tongue are supplied by the hypoglossal nerve *except*:

- (A) Palatoglossus
- (B) Styloglossus
- (C) Hyoglossus
- (D) Genioglossus

57. Meckel's diverticulum is due to persistence of:

- (A) Allantois
- (B) Vitello-intestinal duct
- (C) Left umbilical vein

(D) Ductus venosus

58. The caecum is supplied by:

- (A) Ileocolic artery
- (B) Middle colic artery
- (C) Left colic artery
- (D) Right colic artery

59. The posterior surface of the head of pancreas is related to all *except*:

- (A) Inferior vena cava
- (B) Aorta
- (C) Terminal parts of renal veins
- (D) Bile duct

60. The notable features on the visceral surface of the liver are all *except*:

- (A) Gallbladder
- (B) Bare area of liver
- (C) Porta hepatis
- (D) Fissure of ligamentum teres hepatis

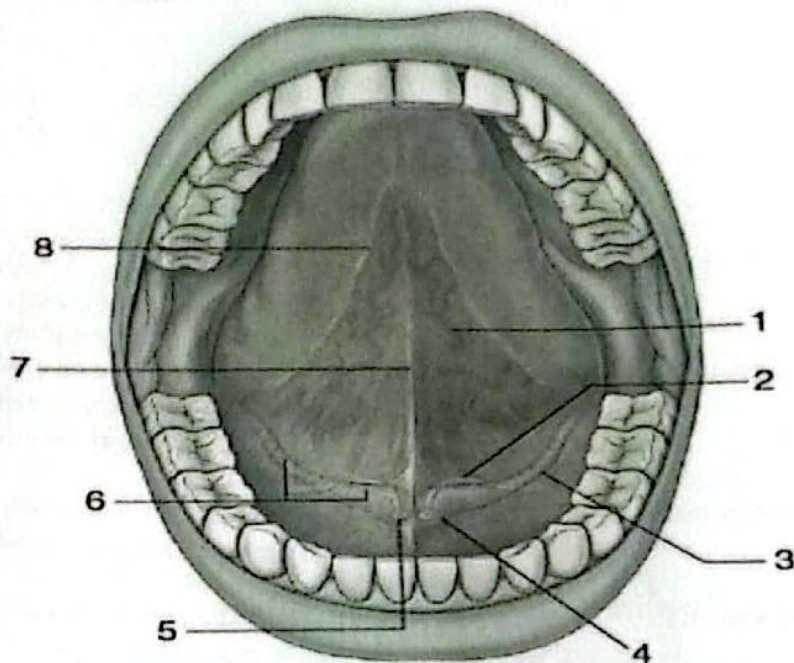
SECTION B

Answer all the questions

1. A Dentist was examining the 3rd right upper molar tooth of a young women, who reported to have been experiencing excruciating pain for a period of time. On examination it was observed that she had a tooth decay and gingiva infection. The solution was tooth extraction.
A. Draw and label all the parts of the tooth (13 mark).
B. Which nerve is the Dentist going to inject anesthetic prior to extraction of the tooth (2 marks)
C. What is the nerve supply to the upper and lower gingiva (4 marks)
2. A foetus born with some abdominal contents protruding outside the abdominal cavity, was been examined by a paediatrician and medical students. There was a debate among the students as to whether it was omphalocele or Gastroschisis.
A. what is the cause of each condition (4 marks)
B. Discuss the two conditions considering their key differences (8 marks)
3. A. Draw and label all the external parts(features) of the Stomach and blood supply (should start from the major artery (14 marks)
4. Describe the special histological structural features that facilitate the processes of digestion & absorption in the small intestines (6 marks)

SECTION C LABEL ALL THE DIAGRAMS A-C
Diagram A

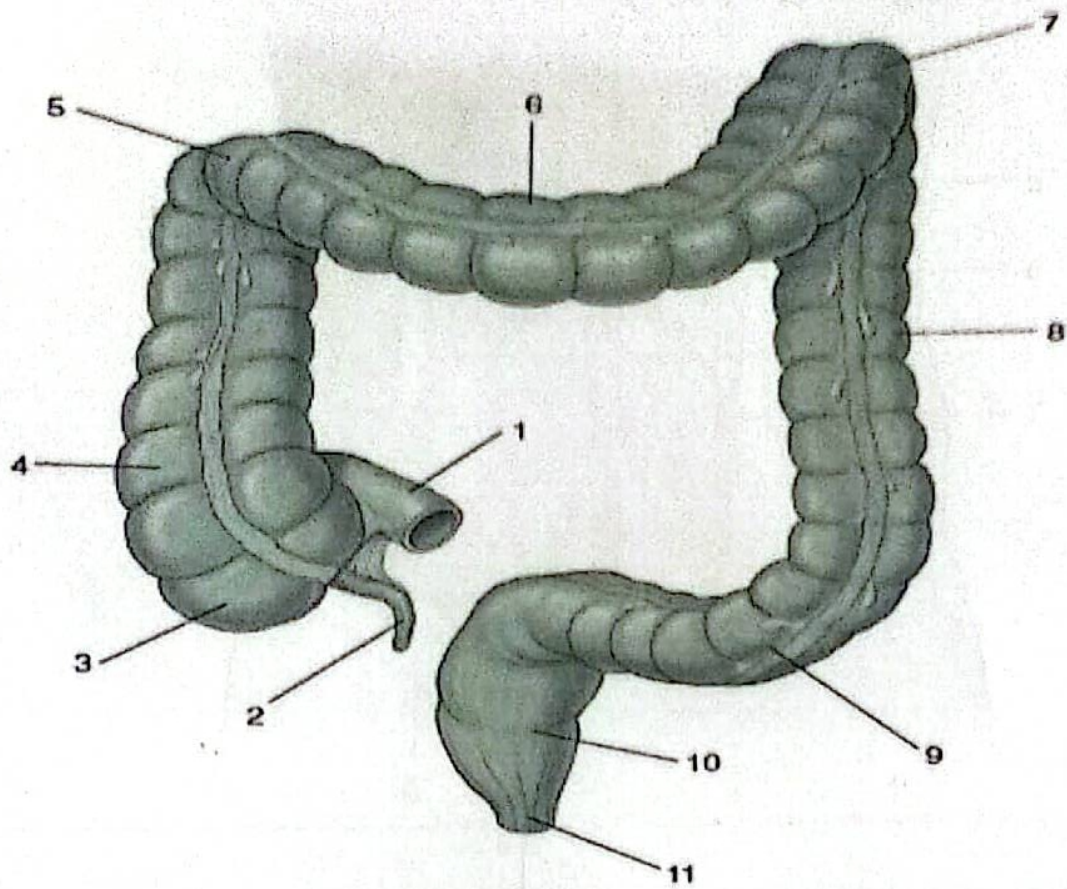
A
Identify the indicated structures.



1	5
2	6
3	7
4	8

Diagram B

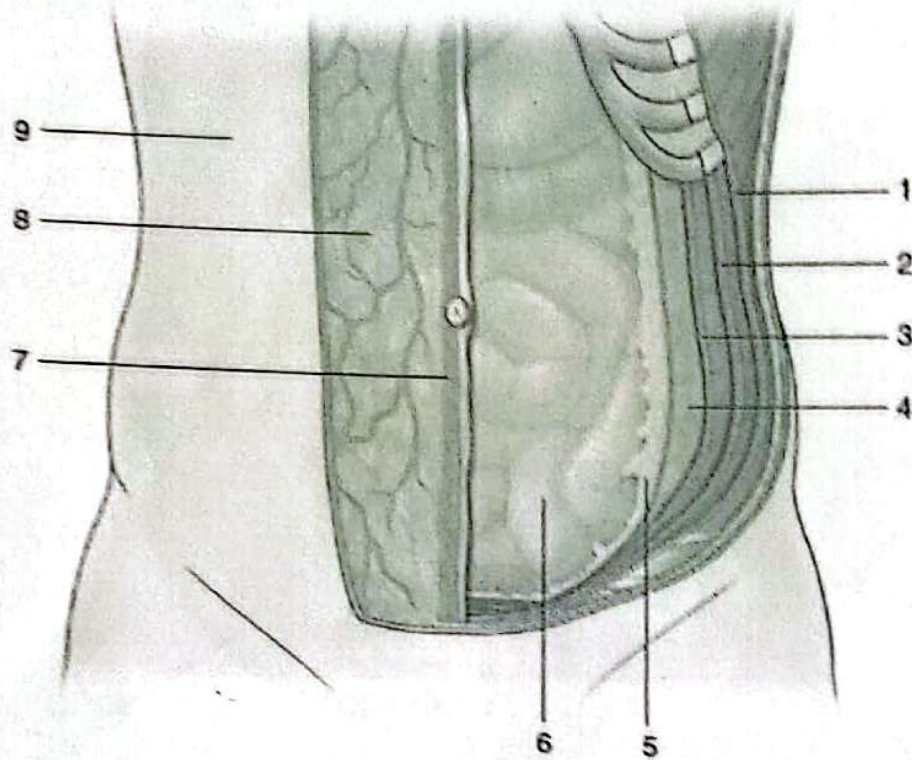
Identify the indicated parts of the large intestine.



1	7
2	8
3	9
4	10
5	11
6	

Diagram C

Identify the indicated layers of the abdominal wall.



1	6
2	7
3	8
4	9
5	10

**MEN IN
FLAMES**



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